

What is an Iterated Block Cipher?

An **Iterated Block Cipher** is a type of encryption method where:

- Plaintext and ciphertext are divided into **fixed-size blocks** (for example, 128 bits).
- The encryption is done by repeating a function called **Round Function (F)** many times.
- This repetition is called **rounds**.

What is Round Function (F)?

- It is called a **round function** not because of its shape,
- But because it is applied repeatedly in multiple rounds.
- Each round makes the data more secure.

What is Feistel Cipher?

- Feistel Cipher is a **structure** used to build block ciphers.
- It is **not a specific algorithm**, but a design method.
- Example: DES uses Feistel structure.

Feistel Cipher (Encryption + Decryption)

Encryption

Take plaintext block and divide into two halves:

$P=(L_0, R_0)$

- L_0 = Left half
- R_0 = Right half

For each round $i = 1$ to n :

$$L_i = R_{i-1}$$

$$R_i = L_{i-1} \oplus F(R_{i-1}, K_i)$$

Where:

- **F** = Round function
- **K_i** = Subkey for that round
- $\oplus$ = XOR operation

After last round: $C=(L_n, R_n)$

That is the ciphertext.

Decryption:

Rounds go backward with ciphertext: $C=(L_n, R_n)$

For round $i = n$ down to 1:

$$R_{i-1} = L_i$$

$$L_{i-1} = R_i \oplus F(R_{i-1}, K_i)$$

Finally we get:

$$P = (L_0, R_0)$$

- Decryption works for **any function F**
- But security depends on choosing a **strong F**

DES Numerology **মানে কী?**

এখানে DES এর গুরুত্বপূর্ণ সংখ্যাগুলো বলা হয়েছে।

★ DES এর প্রধান বৈশিষ্ট্য

১. Feistel Cipher

Data Encryption Standard (DES) একটি Feistel structure ব্যবহার করে।

২. ৬৪ বিট ব্লক

- DES একবারে ৬৪ বিট ডেটা encrypt করে।

৩. ৫৬ বিট key

- আসল security key হলো ৫৬ বিট।
- যদিও key দেখতে ৬৪ বিট, কিন্তু ৮ বিট parity হিসেবে ব্যবহৃত হয়।

৪. ১৬ টি round

- DES মোট ১৬ বার round চালায়।
- প্রতিবার ডেটা আরও বেশি মিশ্র হয়।

৫. প্রতি round এ ৪৮ বিট subkey

- ৫৬ বিট মূল key থেকে প্রতি round এ ৪৮ বিট subkey তৈরি হয়।
- সেটাই round function এ ব্যবহার হয়।

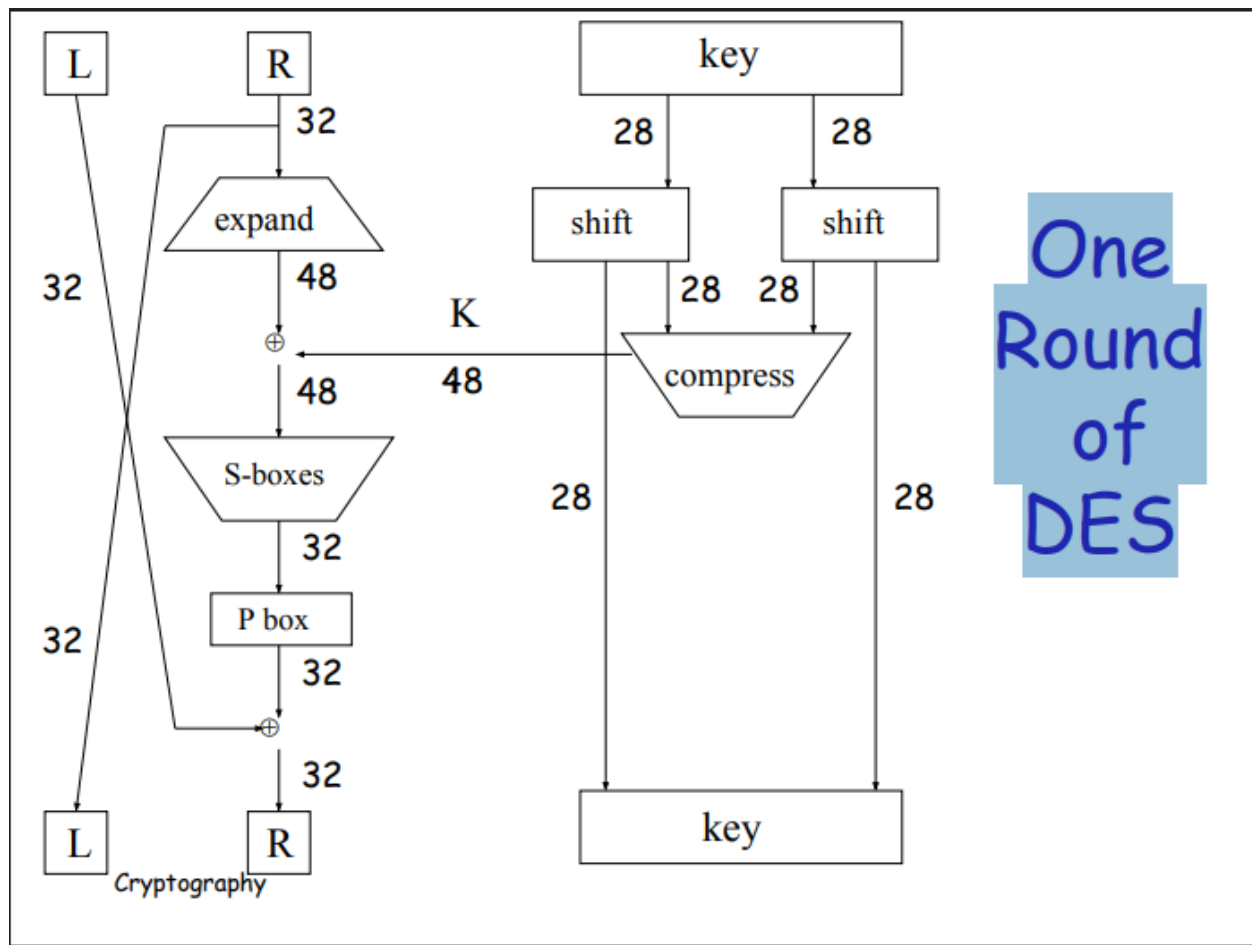
🔒 Security কার উপর নির্ভর করে?

👉 সবচেয়ে গুরুত্বপূর্ণ অংশ হলো S-box।

- প্রতিটি S-box:
 - ৬ বিট input নেয়
 - ৪ বিট output দেয়

এই S-box গুলোই আসল security তৈরি করে।

One Round of DES Block Diagram:



Question and Answer:

Describe the structure and operation of the DES algorithm, including its key size and block size.

Structure of DES

- DES is a **Feistel cipher**.
- **Feistel structure** means:
 1. Plaintext block is split into **left (L) and right (R) halves**.
 2. The right half goes through a **round function (F)**, then combined with the left half using XOR.
 3. Halves are swapped each round.
 4. This is repeated for **16 rounds**.
- There is an **initial permutation (IP)** at the start and a **final permutation (IP⁻¹)** at the end.

Key Size and Block Size

- **Block size:** 64 bits (plaintext divided into 64-bit blocks)
- **Key size:** 56 bits (even though represented as 64 bits, 8 bits are parity)
- **Rounds:** 16 rounds
- **Subkey per round:** 48 bits generated from 56-bit key using **key schedule**.

What role do S-boxes play in block ciphers like DES?

Answer:

In DES, an S-box (Substitution box) is a small lookup table used in the round function.

Role of S-Boxes

1. **Introduce Non-Linearity (Confusion)**
 - Without S-boxes, DES would be mostly linear (XOR, permutation), which is easy to break.
 - S-boxes make the relationship between **key** and **ciphertext** complex.
2. **Mix Input Bits**
 - 6-bit input → 4-bit output
 - Bits from input are **scrambled** in a non-linear way.
3. **Prevent Cryptanalysis**
 - S-boxes are designed to resist **linear and differential attacks**.
 - They are the **main source of security** in DES.
4. **Compact Transformation**
 - Reduces 6-bit input to 4-bit output while adding complexity.
 - Helps the 48-bit expanded R half be compressed back to 32 bits after the round function.

Why DES Encrypt-Decrypt-Encrypt with 2 keys?

Answer:

DES uses Encrypt-Decrypt-Encrypt with 2 keys to increase security (from 56-bit to 112-bit), maintain backward compatibility with single-key DES, and prevent attacks that simple double encryption (E-E) would be vulnerable to.

Bangla:

DES 2টি key ব্যবহার করে Encrypt-Decrypt-Encrypt করে যাতে নিরাপত্তা বৃদ্ধি পায় (56-bit থেকে 112-bit), একক key DES এর সাথে backward compatibility বজায় থাকে, এবং সাধারণ double encryption (E-E) দ্বারা হওয়া আক্রমণ থেকে প্রতিরোধ করা যায়।

❑ In DES cipher.

- a. How many bits in each plaintext block?
- b. How many bits in each ciphertext block?
- c. How many bits in the key?
- d. How many bits in each subkey?
- e. How many rounds?
- f. How many S-boxes?
- g. An S-box requires how many bits of input?
- h. An S-box generates how many bits of output?

a. How many bits in each plaintext block?

- **Answer:** 64 bits
- Explanation: DES encrypts data **64-bit blocks** at a time.

b. How many bits in each ciphertext block?

- **Answer:** 64 bits
- Explanation: DES preserves block size; ciphertext block = 64 bits.

c. How many bits in the key?

- **Answer:** 56 bits
- Explanation: Original key is represented as 64 bits, কিন্তু 8 bits parity হিসেবে থাকে। Security key = 56 bits.

d. How many bits in each subkey?

- **Answer:** 48 bits
- Explanation: প্রতিটি round এর জন্য subkey 48 bits তৈরি হয় মূল 56-bit key থেকে।

e. How many rounds?

- **Answer:** 16 rounds
- Explanation: DES uses **16 Feistel rounds**.

f. How many S-boxes?

- **Answer:** 8 S-boxes per round
- Explanation: 48-bit input → 8 groups of 6-bit → 8 S-boxes process them.

g. An S-box requires how many bits of input?

- **Answer:** 6 bits

- Explanation: Each S-box takes **6-bit input** to produce output.

h. An S-box generates how many bits of output?

- **Answer:** 4 bits
- Explanation: Each S-box produces **4-bit output**, which are combined to form 32-bit output after S-box stage.